

The Hilbertian Product-Measure Subcase of a Question on Ramsey-Type Sphere Constants

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Status

This is a partial solution packet for arXiv:2308.07668, “On Ramsey-type properties of the distance in nonseparable spheres” by Piotr Koszmider. It resolves the case $p = 2$ of the question, asked after Proposition 55, concerning the exact values of the constants c_n , c , and whether the unit sphere of $L_p(\{0, 1\}^\kappa)$ is dichotomous.

Source Evidence

The following crop is from page 25 of the source PDF.

and $\varphi_B \sim \varphi_{A \cap B}$ on $A \cap B$ as the functions enumerating A and B in the increasing order agree on $A \cap B$ as they agree with e_I . So $\|f - f'\|_{L_p}$ is not bigger than c_n by the definition of c_n , as required. $\square$

We do not know the exact value of c_n or of c in the above proof or if S_{L_p} is dichotomous. Note that $k^+(S_{L_p})$ is not $\sqrt[p]{2}$ in general because we can construct in $L_p(\{0, 1\}^{\omega_1})$ uncountable sets of independent variables assuming values 1 and -1 with equal probability $1/2$. They have norm 1 but their differences have the norm equal to $2^{1-1/p}$ which is strictly bigger than $\sqrt[p]{2}$ if p is sufficiently large.

The relevant source sentence is:

“We do not know the exact value of c_n or of c in the above proof or if S_{L_p} is dichotomous.”

Definitions and Source Notation

Let μ be the product probability measure on $\{0, 1\}^\kappa$, where $\kappa \leq 2^\omega$ is uncountable. In Proposition 55 of the source paper, for $A \in [\kappa]^n$, the map T_A sends a function $F \in L_p(\{0, 1\}^n)$ to the function on $\{0, 1\}^\kappa$ depending on the coordinates in A through the increasing enumeration of A . The constant c_n is the supremum of

$$\|T_A(F) - T_B(F)\|_{L_p}$$

over all unit vectors $F \in L_p(\{0, 1\}^n)$ and pairs $A, B \in [\kappa]^n$ whose common coordinates occur in the same positions. Finally $c = \sup_n c_n$.

Proof Intuition

For $p = 2$, the expression defining c_n is a Hilbert-space variance computation. Two copies $T_A(F)$ and $T_B(F)$ share precisely the coordinates in $A \cap B$. Conditional on the shared coordinates, the two remaining parts are independent copies of the same conditional random variable. Therefore their covariance is the square of a conditional expectation, hence nonnegative. The squared distance is consequently at most $2\|F\|_2^2$. Disjoint A and B , together with a mean-zero norm-one F , attain equality.

Lemma 1. *For every $n \geq 1$, the constant c_n from Proposition 55 of the source paper satisfies $c_n = \sqrt{2}$ when $p = 2$.*

Proof. Fix $n \geq 1$, $F \in L_2(\{0, 1\}^n)$ with $\|F\|_2 = 1$, and admissible $A, B \in [\kappa]^n$. Let $I \subseteq \{1, \dots, n\}$ be the set of positions corresponding to $A \cap B$. Equivalently, the random variables $T_A(F)$ and $T_B(F)$ have the joint distribution of $F(X)$ and $F(Y)$, where $X, Y \in \{0, 1\}^n$ are uniform, $X_i = Y_i$ for $i \in I$, and all non-shared coordinates are conditionally independent.

Let $\mathcal{G}$ be the sigma-algebra generated by the shared coordinates. Conditional on $\mathcal{G}$, $F(X)$ and $F(Y)$ are independent copies with the same conditional mean $\mathbb{E}(F(X) \mid \mathcal{G})$. Hence

$$\mathbb{E}(F(X)F(Y) \mid \mathcal{G}) = \mathbb{E}(F(X) \mid \mathcal{G})\mathbb{E}(F(Y) \mid \mathcal{G}) = \mathbb{E}(F(X) \mid \mathcal{G})^2.$$

Thus

$$\begin{aligned} \|T_A(F) - T_B(F)\|_2^2 &= \mathbb{E}|F(X) - F(Y)|^2 \\ &= 2\mathbb{E}|F(X)|^2 - 2\mathbb{E}(F(X)F(Y)) \\ &= 2 - 2\mathbb{E}(\mathbb{E}(F(X) \mid \mathcal{G})^2) \\ &\leq 2. \end{aligned}$$

This proves $c_n \leq \sqrt{2}$.

For the reverse inequality, choose disjoint $A, B \in [\kappa]^n$, possible because κ is infinite and n is finite. Let $F(x) = 1$ if $x_1 = 1$ and $F(x) = -1$ if $x_1 = 0$. Then $\|F\|_2 = 1$ and $\mathbb{E}F = 0$. Since $A \cap B = \emptyset$, $T_A(F)$ and $T_B(F)$ are independent mean-zero norm-one random variables, so

$$\|T_A(F) - T_B(F)\|_2^2 = \|T_A(F)\|_2^2 + \|T_B(F)\|_2^2 - 2\mathbb{E}(T_A(F)T_B(F)) = 2.$$

Therefore $c_n \geq \sqrt{2}$, and equality follows. □

Theorem 1. *For $p = 2$ in Proposition 55 of the source paper,*

$$c_n = \sqrt{2} \quad (n \geq 1), \quad c = \sqrt{2}.$$

Moreover, for uncountable $\kappa \leq 2^\omega$, the unit sphere of $L_2(\{0, 1\}^\kappa, \mu)$ is almost dichotomous but not dichotomous.

Proof. The formula for c_n is Lemma 1, and $c = \sup_n c_n = \sqrt{2}$.

It remains to record the dichotomy status in the same subcase. The Walsh characters indexed by finite subsets of κ form an orthonormal basis of $L_2(\{0, 1\}^\kappa, \mu)$. Since $|\kappa|^{<\omega} = \kappa$, this Hilbert space is linearly isometric to $\ell_2(\kappa)$. Proposition 54 of the source paper, applied with $p = 2$, states that

$$\mathbf{K}^+(S_{\ell_2(\kappa)}) = [0, \sqrt{2}), \quad \Sigma(S_{\ell_2(\kappa)}) = (\sqrt{2}, 2].$$

Consequently $S_{L_2(\mu)}$ is almost dichotomous and not dichotomous. □

Verification Notes

The proof of $c_n \leq \sqrt{2}$ is a direct conditional-expectation identity. Equality is attained by disjoint coordinate sets and a Rademacher function on one coordinate. The auxiliary script in the packet's `code/` directory builds the finite quadratic forms for $1 \leq n \leq 5$ and confirms that the largest squared distance is 2 for every overlap pattern; this is only a finite sanity check and is not used as evidence in place of the proof.

Command used:

```
conda run --no-capture-output -n sandbox python code/check_l2_constant.py
```

Scope and Limitations

This packet does not determine c_n , c , or dichotomy for $p \neq 2$. The non-dichotomy statement for $p = 2$ is obtained by identifying $L_2(\mu)$ with $\ell_2(\kappa)$ and invoking the source paper's earlier Hilbert-space result. The packet's direct computation is the exact value of the product-measure constants c_n in the Hilbertian case.

Novelty and Search Bounds

The run indexes were searched for arXiv:2308.07668, the paper title, and L_p/L_2 keywords, with no duplicate packet found. A bounded web phrase search for the exact source sentence and nearby terms returned no obvious later answer. Novelty confidence is modest: this is an elementary Hilbertian subcase adjacent to results already present in the source paper, not a claimed resolution of the full $1 < p < \infty$ problem.

Human Review Recommendation

Reviewers should check the conditional-independence model for the allowed pairs A, B , the equality case using disjoint coordinate sets, and the Hilbert-space identification via the Walsh basis. The result should be counted only as a partial subcase.

References

- [1] P. Koszmider, *On Ramsey-type properties of the distance in nonseparable spheres*, arXiv:2308.07668.